



İSTANBUL TOPKAPI UNIVERSITY

FACULTY OF ENGINEERING AND NATURAL SCIENCES

27.10.2025

To whom it may concern,

We have reviewed the study titled “Evaluating Pharmaceutical Reverse Logistics Sustainability with a Pythagorean Fuzzy Multi-Criteria Approach” conducted by Tolga Kudret Karaca. The study is non-interventional and limited to the elicitation of judgments from professional experts regarding evaluation criteria. No personal or health data were collected. No identifying information was recorded. No experimental procedures involving human participants were performed.

Under our institutional policy and applicable regulations, this type of activity is **exempt from ethics committee review**. Accordingly, **ethical approval is not required** for this research.

Please do not hesitate to contact us if further information is required.

Sincerely,
Prof. Dr. Onur OSMAN

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